C. Preservation by drying
   1. Sun drying
   2. Drying by mechanical driers
      (i) Spray drying
      (ii) Foam mat drying
      (iii) Drying by osmosis
      (iv) Freeze drying
D. Preservation by irradiation

II. Chemical Methods

A. High concentration of salt
B. High concentration of sugar
C. Using chemical preservatives

III. Fermentation

6.2.1 Physical Methods

A. Preservation by Low Temperatures

1. Refrigeration: The temperature maintained in the refrigerator is 0 to 5°C. Enzymatic and microbial changes in foods are slowed down considerably. Perishable foods like eggs, dairy products, meat, sea foods, fruits and vegetables are stored in refrigerators. Food can be stored safely for few days or a week.

2. Freezing: The temperature of freezer is –18 to –40°C. Microbial growth is prevented completely and the action of food enzymes greatly reduced. Frozen foods have better quality and needs uninterrupted supply of electricity while storing. Foods like poultry, meat, fish, peas, vegetables, juice concentrates can be preserved for several months by this method. In vegetables, enzyme action may still produce undesirable effects on flavour and texture during freezing. Various methods of freezing include slow freezing, quick freezing, cryogenic freezing and dehydro freezing.

Quick freezing is an ideal method of preserving nearly all baked products. Bread frozen at –22°C or below retains its freshness for many months. Cakes, cookies, short cakes, waffles and pancakes are also frozen and marketed frozen.

B. Preservation by High Temperature

Food commodities in which pasteurization is employed include milk, beer, wines and fruit juices.

1. Pasteurization: Pasteurization is a mild heat treatment that kills a part but not all



▲ Plate 6.1 Pasteurization Treatment

the microorganisms present and usually involves the application of temperature below 100°C.

There are three methods of pasteurization.

a. **Bottle or Holding Pasteurization:** This method is commonly used for the preservation of fruit juices. The extracted juice is strained and filled in bottles. The bottles are then sealed air tight and pasteurized.

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